The empirical recurrence for OEIS A262468: recovering a conjecture that is not written down

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Abstract

OEIS A262468 records "Empirical recurrence of order 73", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 525$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 73, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 73 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A262468 is "Number of (n+1)X(4+1) 0..1 arrays with each row divisible by 3 and each column divisible by 5, read as a binary number with top and left being the most significant bits.". It has offset 1 and begins

$$1, 11, 121, 457, 2413, 22907, 239281, 2028469, \ldots$$

The entry states:

> Empirical recurrence of order 73 (see link above)

As of the "Last modified" line on the live entry (Sep 23 2015 18:49:17, revision 4) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 3125$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 525$ of the 3125, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 525$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 1050$ exact terms of the model returns order 73 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{73} c_i \, a(n-i),$$

c_1	11	c_{20}	503037392585749968	c_{39}	0
c_2	0	c_{21}	−5533411318443249648	c_{40}	−3318443441220387490307522659308
c_3	0	c_{22}	0	c_{41}	36502877853424262393382749252388
c_4	16473	c_{23}	0	c_{42}	0
c_5	−181203	c_{24}	−383859517771004581604	c_{43}	0
c_6	0	c_{25}	4222454695481050397644	c_{44}	397322327858143836927875847520164
c_7	0	c_{26}	0	c_{45}	−437054560643958220620663432272180
c_8	−90791988	c_{27}	0	c_{46}	0
c_9	998711868	c_{28}	202296232636635472704204	c_{47}	0
c_{10}	0	c_{29}	−2225258559002990199746244	c_{48}	−317742465777500207335225597543949
c_{11}	0	c_{30}	0	c_{49}	34951671235552502280687481572983443
c_{12}	258835986796	c_{31}	0	c_{50}	0
c_{13}	−2847195854756	c_{32}	−74144401341923006671040898	c_{51}	0
c_{14}	0	c_{33}	815588414761153073381449878	c_{52}	16503278898738624820545218333201884
c_{15}	0	c_{34}	0	c_{53}	−1815360678861248730259974016652207
c_{16}	−446777695281960	c_{35}	0	c_{54}	0
c_{17}	4914554648101560	c_{36}	18882281235364654443508906170	c_{55}	0
c_{18}	0	c_{37}	−207705093589011198878597967870	c_{56}	−5357844957959520080649421081004390
c_{19}	0	c_{38}	0	c_{57}	589362945375547208871436318910482905

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 73, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $73 + 4$ terms removed returns order 73 as well, so $\deg q_{\min} = 73 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. ☐

4 Verification

Three checks.

First, the walk model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $73 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 73 that this sequence satisfies — and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A262468.

[2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.